Sahaj Bhadja

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Summary

Associate-level engineer with 2+ years building **SQL/Python analytics workflows**, **cloud data integrations**, and **customer-facing product features**, with strong communication across technical and business stakeholders.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*

- Built **data-facing backend services** and interfaces for 500+ users, reducing latency from **1.2s to 350ms** via SQL optimization and improving reliability of dashboard-driven analytics and stakeholder decision workflows.
- Implemented **FastAPI-based ETL orchestration** for 5K+ monthly jobs with validation, reconciliation, and issue logging, producing trustworthy outputs and clear technical artifacts for cross-functional and non-technical collaborators.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*

- Developed **React + Flask product capabilities** handling 10K+ daily requests, translating user requirements into APIs, workflows, and explainable outputs while collaborating directly with product and engineering stakeholders.
- Improved release quality through CI/CD checks and automated validation, reducing turnaround by **60%** and enabling faster iterations for customer-facing improvements and internal demo-ready feature rollouts.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*

- Engineered **API and ETL services** with Python/Java, AWS Glue, and S3, improving response times by **30%** while supporting scalable cloud data movement and downstream reporting integrations.
- Maintained **20+ automated unit/integration suites** with Jenkins, PyTest, and Mockito, boosting CI efficiency by **68%** and reducing technical debt by **20%** in Agile cross-team delivery.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*

- Created Python analysis pipelines over 100K+ trajectories, generating concise summaries and diagnostics that helped technical and non-technical partners interpret model behavior and prioritize validation tasks.
- Presented findings and implementation recommendations to multidisciplinary teams, improving collaboration quality and accelerating transition from exploratory analysis to production-ready engineering plans and milestone decisions.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*

- Built REST/WebSocket modules supporting 2M+ synchronized records and eight workflows, improving reliability of operational updates and data consistency across customer-facing interfaces and backend processing components.
- Automated Docker/Jenkins deployment and test workflows, reducing release time by **50%** and improving reproducibility of technical demonstrations, bug fixes, and iterative software delivery.

Skills

Data & Analytics: **SQL**, SQL analytics, Python analysis tooling, ETL workflows, data quality validation
Cloud & Platforms: AWS (S3/Glue/RDS), Azure SQL, containerized services, CI/CD integration
Application Engineering: React.js, Flask, FastAPI, REST APIs, microservices, technical documentation
Collaboration: customer problem framing, technical presentations, cross-functional solution design, iterative POC delivery

Projects

Chat2SQL Code | *Java, TypeScript, SQL, Docker*

- Built a schema-aware SQL workspace across 10+ databases with validated query generation and structured outputs, enabling faster data exploration, clearer stakeholder communication, and stronger confidence in analytical decision-making.

Linguistix | *Python, FastAPI, AWS ECS/Fargate, CodePipeline*

- Developed a containerized API with automated quality checks and cloud deployment controls, achieving **87%** processing accuracy while supporting repeatable demos, reliable releases, and maintainable data-service operations.

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB*

- Engineered a full-stack fintech platform supporting 50K+ users and 1M+ transactions, combining secure APIs, real-time synchronization, and reporting-friendly workflows for robust financial product experiences and customer insights.